THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	Each stage triples the count: $3^1 = 3$
2	**(d)** 21	Sum $= 105$; $\div 5$
3	**(c)** $2x + 5$	2 times x is $2x$; add 5
4	**(a)** 84 cm²	12×7
5	**(c)** 5	$n + 2 = 5$ (two bases and n rectangles)
6	**(c)** 28	Arrange in order; take the middle value
7	**(a)** 19	$3 \times 5 + 4$
8	**(c)** 30 cm²	$\frac{1}{2} \times 10 \times 6$
9	**(d)** 4	n base vertices + 1 apex $= 4$
10	**(b)** 18	Most frequent value
11	**(b)** $6x + 8$	Add like terms
12	**(d)** 36 cm²	base × height $= 9 \times 4$
13	**(d)** 9	$3n = 9$
14	**(a)** 33	Highest − lowest
15	**(a)** $10x + 2$	Place value: tens × 10 + ones

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Faces $= n + 2 = 5$ (1); vertices $= 2n = 6$, edges $= 3n = 9$ (1).	1+1
17	Sum $= 140$ (1); mean $= 140 \div 5 = 28$ (1).	1+1
18	$5n + 7$ (2).	2
19	Area $= l \times b = 25 \times 12$ (1) $= 300$ m² (1).	1+1
20	$F + V - E = 6 + 8 - 12$ (1) $= 2$, so Euler's formula is verified (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Faces $= 4$, vertices $= 4$, edges $= 6$ (2); $F + V - E = 4 + 4 - 6 = 2$ (1).	2+1
22	Mean $= 270 \div 6 = 45$ (2); students above mean $= 2$ (1).	2+1
23	$3 \times 6 + 4 \times 2 - 5$ (2) $= 21$ (1).	2+1
24	Area $= bh = 126$ cm² (2); triangle $= \frac{1}{2} \times 126 = 63$ cm² (1).	2+1
	OR: Stage 2: $3^2 = 9$ (1); stage 3: $3^3 = 27$ (1); more $= 18$ (1).	1+1+1
25	Ascending: 6, 7, 21, 37, 44, 50, 55, 69, 70 (1); 9 values, middle one is the 5th (1); median $= 44$ (1).	1+1+1
	OR: Like terms: $9x - 5x = 4x$ (1); constants $4 + 8 - 2 = 10$ (1); result $4x + 10$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Stage 1: 4 segments of 3 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 1$ cm (1); total $= 16 \times 1 = 16$ cm (2).	1+1+1+2
	OR: Mean $= 15.17$ (1); median $= 15.5$ (2); mode $= 25$ (1); range $= 21$ (1).	1+2+1+1
27	$3n + 5 - 5 = 3n$ (3); for $n = 4$, result $= 12$ (2).	3+2
	OR: Side $= 24 \div 4 = 6$ m (1); area $= 6^2 = 36$ m² (2); tiles $= 36$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 35 cm²	$\frac{1}{2}(8+6)\times 5$
2	**(c)** 12	$F+V-E=2$ gives $E=6+8-2=12$
3	**(b)** 90	Sum = mean × number of values
4	**(a)** 28	$3+4+10+11$
5	**(c)** 40 cm²	$\frac{1}{2}\times 10\times 8$
6	**(b)** 36	A cube has 12 edges: $12\times 3=36$
7	**(d)** one observation	One dot per data value
8	**(d)** 5	Number multiplying the variable
9	**(b)** 25 cm²	Side $=5$ cm; area $=5^2$
10	**(b)** $F+V-E=2$	Euler's formula
11	**(b)** 13	Sum $=65$; $\div 5$
12	**(a)** $3x+4$	3 times x is $3x$; add 4
13	**(b)** 120 cm²	15×8
14	**(a)** 9	Each stage triples the count: $3^2=9$
15	**(c)** 24	Arrange in order; take the middle value

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Ascending order: 5, 32, 33, 38, 53, 56, 60 (1); median $=38$ (1).	1+1
17	$3\times 4+2\times 5=22$ (1+1).	1+1
18	Area $=\frac{1}{2}bh=\frac{1}{2}\times 18\times 10$ (1) $=90$ cm² (1).	1+1
19	A cube has 12 equal edges (1); total $=12\times 4=48$ cm (1).	1+1
20	Range $=15-9=6$ (1); mode $=10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Area $=\frac{1}{2}(a+b)h=\frac{1}{2}(13+7)\times 9$ (2) $=90$ cm² (1).	2+1
22	A cuboid has 4 edges of each length (1); total $=4(5+3+2)=40$ cm (1); faces 6, vertices 8 (1).	1+1+1
23	Rises: 4, 7, 3 (2); greatest rise 7°C (1).	2+1
24	$23+32=55$ (2); $=11\times 5$, a multiple of 11 (1).	2+1
	OR: Area $=8\times 5=40$ m² (2); cost $=40\times 40=$ ₹ 1600 (1).	2+1
25	Faces $=5$, vertices $=5$, edges $=8$ (2); $F+V-E=5+5-8=2$ (1).	2+1
	OR: Mean $=318\div 6=53$ (2); students above mean $=3$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Stage 1: 3 (1); stage 2: 9 (1); stage 3: 27 (1); each stage multiplies the number by 3 (2).	1+1+1+2
	OR: Sum of five $=5\times 20=100$ (2); sum of four $=85$ (1); fifth $=15$ (2).	2+1+2
27	Middle row: 2+5=7 (1), 5+3=8 (1); top $=7+8=15$ (3).	1+1+3
	OR: Area $=\frac{1}{2}\times 16\times 12=96$ cm² (3); each triangle $=\frac{1}{4}$ of it $=24$ cm² (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 19	$2 \times 6 + 7$
2	**(a)** 42 cm²	$\frac{1}{2} \times 12 \times 7$
3	**(b)** 7	$n + 2 = 7$ (two bases and n rectangles)
4	**(b)** 7	Most frequent value
5	**(a)** $8x + 8$	Add like terms
6	**(d)** 66 cm²	base × height $= 11 \times 6$
7	**(b)** 6	n base vertices + 1 apex $= 6$
8	**(b)** 44	Highest − lowest
9	**(d)** $10x + 3$	Place value: tens × 10 + ones
10	**(d)** 42 cm²	$\frac{1}{2}(10 + 4) \times 6$
11	**(d)** 15	$3n = 15$
12	**(a)** 96	Sum = mean × number of values
13	**(c)** 40	$6 + 7 + 13 + 14$
14	**(c)** 54 cm²	$\frac{1}{2} \times 12 \times 9$
15	**(a)** 8	$F + V - E = 2$ gives $E = 5 + 5 - 2 = 8$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$7x + 7y$ (1+1).	1+1
17	Area $= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 22 \times 10$ (1) $= 110$ cm² (1).	1+1
18	Faces $= n + 2 = 7$ (1); vertices $= 2n = 10$, edges $= 3n = 15$ (1).	1+1
19	Sum $= 190$ (1); mean $= 190 \div 5 = 38$ (1).	1+1
20	$4y - 9$ (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$5 \times 4 + 2 \times 7 - 8$ (2) $= 26$ (1).	2+1
22	Area $= bh = 112$ cm² (2); triangle $= \frac{1}{2} \times 112 = 56$ cm² (1).	2+1
23	Stage 3: $3^3 = 27$ (1); stage 4: $3^4 = 81$ (1); more $= 54$ (1).	1+1+1
24	Ascending: 21, 28, 29, 45, 47, 54, 55, 58, 65 (1); 9 values, middle one is the 5th (1); median $= 47$ (1).	1+1+1
	OR: Like terms: $12x - 7x = 5x$ (1); constants $3 + 6 - 4 = 5$ (1); result $5x + 5$ (1).	1+1+1
25	Area $= \frac{1}{2}(a + b)h = \frac{1}{2}(16 + 10) \times 6$ (2) $= 78$ cm² (1).	2+1
	OR: A cuboid has 4 edges of each length (1); total $= 4(8 + 4 + 3) = 60$ cm (1); faces 6, vertices 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Stage 1: 4 segments of 6 cm (1); stage 2: $4 \times 4 = 16$ segments (1); each of length $= 2$ cm (1); total $= 16 \times 2 = 32$ cm (2).	1+1+1+2
	OR: Mean $= 13.5$ (1); median $= 10.5$ (2); mode $= 7$ (1); range $= 21$ (1).	1+2+1+1
27	$4n + 7 - 7 = 4n$ (3); for $n = 6$, result $= 24$ (2).	3+2
	OR: Side $= 32 \div 4 = 8$ m (1); area $= 8^2 = 64$ m² (2); tiles $= 64$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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